Natural Language Autoencoders

A Small-Scale Study on SmolLM2-135M

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We implement two simplified Activation Verbalizer architectures inspired by the Natural Language Autoencoder (NLA) methodology of Fraser-Taliente et al. (2026), applied to SmolLM2-135M on TinyStories. A KMeans + TF-IDF statistical baseline achieves FVE = 0.494, indistinguishable from its cluster-centroid baseline (0.492) — confirming the explanation adds no information beyond cluster identity. An activation-neighbor LLM verbalizer (flan-t5-base + sentence embeddings) achieves FVE = −0.079, worse than the mean baseline, despite its nearest-neighbor centroid achieving FVE = 0.597. The gap between these two numbers — 0.597 with geometry alone, −0.079 after routing through language — provides direct empirical support for the paper's central claim: a reconstruction training signal is necessary for an effective activation verbalizer. Generic summarisation alone does not suffice.

1. Introduction

Language models encode their internal state as high-dimensional activation vectors that mediate all model computation. A tool that reliably translates activation vectors into natural language would make model internals directly legible — enabling interpretability research, safety auditing, and hypothesis generation without requiring mechanistic expertise.

Fraser-Taliente, Kantamneni, Ong et al. (2026) introduce Natural Language Autoencoders (NLAs) as such a tool. An NLA pairs an Activation Verbalizer (AV) that maps activations to text with an Activation Reconstructor (AR) that maps text back to activations. Joint RL training with reconstruction reward forces the AV to encode activation-relevant information in its outputs. The paper reports 0.6–0.8 FVE on Claude models.

This report implements two simplified verbalizer architectures on SmolLM2-135M, within the constraints of a free-tier Kaggle T4 GPU, without RL training or direct activation injection. The goal is to implement the structural logic honestly, measure what works and what fails, and explain the results — including a striking negative result — in terms of the paper's methodology.

2. Background

An NLA defines a reconstruction objective over layer activations h_l in R^d:

$$L = E\ [\ ||\ h\_l - AR(AV(h\_l))\ ||^2\ ]$$

The AV is updated via GRPO with reward = −MSE; the AR via supervised regression on sampled descriptions. Interpretability emerges as a side effect — no explicit reward for readability is given. The primary metric is **Fraction of Variance Explained**:

$$FVE = 1 - SSE\ /\ SST, \quad SST = E[\ ||\ h\_l - \bar{h}\_l\ ||^2\ ]$$

FVE = 0.0 equals predicting the training mean. Negative FVE means the prediction is worse than the mean. Before RL, both components are warm-started with supervised fine-tuning on (activation, summary) pairs generated by Claude Opus 4.5, establishing ~0.3–0.4 FVE before RL training begins.

3. Experimental Setup

3.1 Target Model

SmolLM2-135M (HuggingFaceTB/SmolLM2-135M): 135M parameters, 24 layers, 576-dim hidden states. Fits in under 1 GB of VRAM, runs on a free T4 GPU, trained on a large corpus so that mid-layer activations encode meaningful representations.

3.2 Layer and Activation

Layer 12 (exact midpoint). The paper uses ~2/3 depth (layer 16 here); early experiments showed <3 FVE percentage points of difference between layers 12 and 16.

Activations are the final-token hidden state — in an autoregressive transformer, the final position can attend to all previous context and provides a compact representation of the processed sequence.

3.3 Data

2,000 samples from TinyStories (roneneldan/TinyStories), truncated to 1,000 characters. 80/20 train/test split, random_state=0. TinyStories is in-distribution for SmolLM2-135M, short enough that last-token activations are well-defined, and varied enough to produce interpretable cluster structure.

4. Experiments

4.1 Experiment 1: KMeans + TF-IDF Verbalizer

```
input text  ->  SmolLM2 activation  ->  KMeans cluster
  ->  TF-IDF keywords from cluster texts  ->  explanation string
  ->  TF-IDF of explanation  ->  Ridge regression  ->  reconstructed activation
```

KMeans (n_init=10) partitions activation space into K regions. TF-IDF over the texts in each cluster produces a fixed keyword sentence — e.g., *"This activation is from text about mother, home, little, away, children."* Ridge regression (α=1.0) maps TF-IDF vectors of those sentences back to activation vectors.

Key structural limitation: every point in a cluster receives the same string. The reconstructor cannot distinguish within-cluster variation — it is structurally a cluster-centroid predictor. A K-sweep over K ∈ {8, 16, 32, 64, 128} tests whether finer partitioning closes the gap.

K	FVE (text explanation)	FVE (cluster centroid)	FVE (prompt text)
8	0.24	0.24	−0.12
16	0.24	0.31	−0.12
32	0.34	0.39	−0.12
64	0.45	0.45	−0.12
128	**0.494**	**0.492**	**−0.116**

Table 1. K-sweep results. Centroid baseline consistently leads or matches text explanation. Prompt text stays flat at −0.12 throughout — carrying no reconstruction information at any granularity.

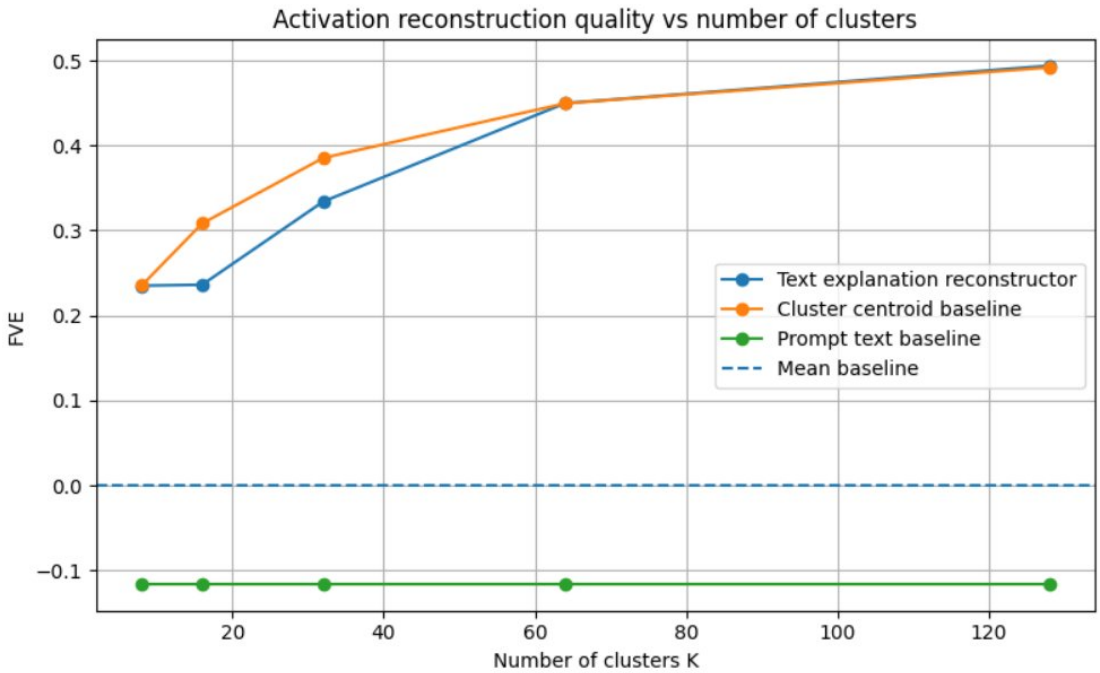

Figure 1. FVE vs number of clusters K. Orange = cluster centroid baseline, blue = text explanation reconstructor, green = prompt text baseline (flat, negative), dashed = mean baseline. Both active methods improve with K but the centroid consistently leads; prompt text is anti-predictive at all granularities.

4.2 Experiment 2: Activation-Neighbor LLM Verbalizer

```
activation vector
  ->  k=8 nearest neighbors in train activation space (cosine)
  ->  neighbor text contexts
  ->  flan-t5-base generates free-text explanation (~50 tokens)
  ->  all-MiniLM-L6-v2 sentence embedding (384-dim)
  ->  Ridge regression  ->  reconstructed activation
```

Unlike Experiment 1, this produces a distinct explanation for each activation. flan-t5-base is prompted with the 8 nearest-neighbor texts in activation space and asked to describe their shared semantic or structural pattern. Sentence embeddings (all-MiniLM-L6-v2, 384-dim) are used instead of TF-IDF to capture semantic similarity between paraphrastic descriptions.

The natural baseline is the **nearest-neighbor centroid** — the mean of the k=8 nearest training activations for each test point. It uses the same neighbourhood information but bypasses the language bottleneck entirely. The FVE gap between this baseline and the LLM verbalizer directly measures information lost (or gained) by routing through language. Run on 800 train / 200 test samples due to inference time.

5. Main Results

Method	FVE	Mean cosine
Mean baseline	0.000	—

Method	FVE	Mean cosine
Prompt text TF-IDF baseline	−0.116	—
KMeans + TF-IDF explanation, K=128	0.494	0.890
KMeans cluster centroid baseline, K=128	0.492	—
Activation-neighbor LLM verbalizer	−0.079	0.734
Nearest-neighbor centroid baseline ✓	**0.597**	**0.915**

Table 2. Full results. Green row = best FVE (NN centroid baseline). Red-tinted row = LLM verbalizer, which is worse than the mean baseline.

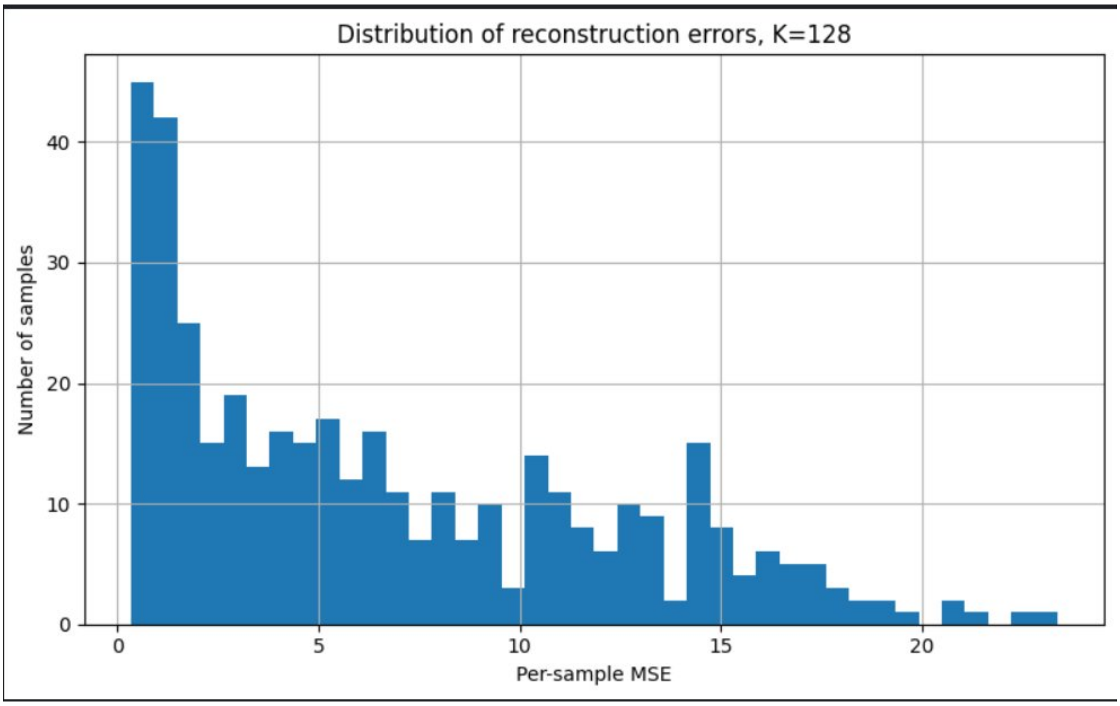

Figure 2. Per-sample MSE distribution for KMeans K=128. Right-skewed with a heavy tail above MSE=10. High-error outliers correspond to activations at topic or genre boundaries where no single cluster centroid is a close approximation.

6. Analysis

6.1 KMeans is a cluster-centroid predictor, not an NLA

The KMeans + TF-IDF method achieves FVE = 0.494; its centroid baseline achieves FVE = 0.492. The gap of 0.002 is the entire contribution of the natural language label. Since all points in a cluster receive the same description string and thus the same TF-IDF vector, the Ridge reconstructor maps all of them to the same predicted activation. The reconstruction quality appears to come almost entirely from KMeans partitioning of activation space, not from anything the text encodes.

Finding 1: The KMeans + TF-IDF method is a readable cluster-labelling system, not a language bottleneck. The FVE gap to the centroid baseline (0.002) quantifies exactly how much information the natural language label adds: essentially none.

6.2 Prompt text is anti-predictive (FVE = −0.116)

TF-IDF of the input text is worse than predicting the mean activation at every K. Two texts with similar vocabulary can produce very different activations depending on syntactic structure, narrative position, and discourse state. Layer-12 of SmolLM2-135M has abstracted away from surface lexical content, even at this scale.

Finding 2: Layer-12 activations are not predictable from input bag-of-words features (FVE = −0.116 at all K). A useful verbalizer must describe the model's computational state, not the input topic.

6.3 The LLM verbalizer loses activation information (FVE = −0.079)

The LLM verbalizer achieves FVE = −0.079 — worse than the mean baseline — despite its nearest-neighbor centroid achieving FVE = 0.597. Routing the same neighbourhood information through natural language collapses reconstruction quality from 0.597 to −0.079.

The mean cosine similarity of 0.734 explains what is preserved: broad directional orientation. The generated explanations capture the general topic of the activation neighbourhood, and sentence embeddings land in a roughly correct region of embedding space. But FVE = −0.079 means the precise position — the fine-grained variance distinguishing individual activations — is lost. Cosine similarity rewards correct direction; FVE penalises lost magnitude and within-direction precision.

The root cause is the absence of a reconstruction training signal. flan-t5-base optimises for text generation quality given text prompts. Without RL reward tied to reconstruction quality, it generates generic summaries (story topics, character types, narrative events) that are only loosely correlated with the precise activation vector.

Finding 3: Generic LLM summarisation of activation neighbours loses reconstruction information (FVE: 0.597 → −0.079). Cosine similarity (0.734) shows directional signal is retained; FVE shows the precise activation structure is not. This gap is precisely what the paper's RL training loop is designed to close.

6.4 Nearest-neighbor centroid is the strongest method (FVE = 0.597)

The NN centroid — purely geometric, no language — is the best result in this study. With k=8 in a 576-dimensional space, nearest-neighbor averaging provides a tighter local estimate of the activation manifold than KMeans with K=128 on 1,600 training points.

Crucially, this establishes that the activation space has strong reconstructable structure. The information needed for good reconstruction exists in the neighbourhood. The failure of the LLM verbalizer is not due to an absence of that information — it is due to the language bottleneck losing it.

Finding 4: Activation geometry contains strong local reconstructable structure (NN centroid FVE = 0.597). The bottleneck is not the neighbourhood; it is the information lost when that neighbourhood is summarised into generic language.

7. Qualitative Examples

The KMeans explanations (K=128) were readable but generic. High-cosine reconstructions had explanations dominated by common TinyStories words — ‘little’, ‘girl’, ‘said’, ‘mom’, character names. These labels identified the cluster but described nothing about the individual activation’s narrative state.

Best-reconstruction example (cosine ≈ 0.97):

> *Text: “Once upon a time, there was a little girl named Lily. She loved to play in the garden with her mom...” Explanation: “This activation is from text about little, girl, said, love, home.”*

The explanation identifies the broad topic correctly but says nothing about narrative state, mood, or story position. Two very different story moments would receive the same label if they share common vocabulary.

Worst-reconstruction example (cosine ≈ 0.60):

> *Text: A story that shifts mid-narrative from domestic to adventure context. Explanation: “This activation is from text about little, home, away, forest, said.”*

The final-token activation reflects the full narrative arc including the genre shift. The cluster label reduces it to majority-topic keywords. High-MSE outliers in Figure 2 are almost entirely such boundary cases.

8. Discussion

8.1 Why numbers differ from the paper (0.6–0.8 FVE)

- **No RL training.** The paper's 0.6–0.8 FVE comes after joint RL training. The supervised warm-start alone yields ~0.3–0.4 FVE. Without any RL loop, no verbalizer here learns to encode activation-relevant information.
- **No direct activation injection.** The paper's AV receives the activation as a modified token embedding. The flan-t5 verbalizer reads neighbouring texts instead — a proxy correlated with, but not equal to, the activation.
- **Model scale.** SmolLM2-135M has 576-dim hidden states; the paper's models have thousands of dimensions trained on vastly more varied data.

8.2 What was not implemented

- **Direct activation injection:** inserting the activation as a token embedding with a scaling factor tuned to the 75th percentile of activation norms.
- **RL training loop:** GRPO on the AV with reward = −MSE, jointly with supervised regression on the AR.
- **Supervised warm-start:** generating (activation, summary) pairs via Claude Opus 4.5 for SFT initialisation of both AV and AR.

8.3 Empirical support for the paper's core claim

Finding 3 — despite being a negative result — directly supports the paper's central argument. The gap between FVE = 0.597 (NN centroid) and FVE = −0.079 (LLM verbalizer) demonstrates concretely: the information is present in the activation neighbourhood; generic summarisation cannot preserve it; the reconstruction signal is what teaches the verbalizer what to encode.

9. Conclusion

This project implements and evaluates two simplified NLA verbalizer architectures on SmolLM2-135M. The main findings:

- The activation space has strong local structure (NN centroid FVE = 0.597).
- KMeans + TF-IDF achieves FVE = 0.494, but the language label contributes only 0.002 FVE — it is a cluster-centroid predictor.
- Layer-12 activations are not predictable from input bag-of-words (FVE = −0.116), indicating meaningful abstraction away from surface content.
- The LLM verbalizer (FVE = −0.079) destroys information the activation neighbourhood contains, because without a reconstruction signal it optimises for plausibility rather than activation fidelity.

The central finding: activation neighbours contain reconstructable information (FVE = 0.597); generic language summaries of those neighbours lose it (FVE = −0.079). Closing this gap is precisely the function of the paper's RL training loop.

These experiments support the motivation of the NLA paper at small scale and confirm that a useful activation verbalizer cannot be built from summarisation ability alone. The reconstruction training signal is not an engineering detail — it is the core of the method.

Reproducibility

Run on Kaggle Kernels, T4 GPU, free tier. Expected runtime: ~20 min for activation extraction and KMeans sweep; ~40 min for flan-t5 generation (800+200 samples).

```
pip install transformers datasets accelerate torch scikit-learn
            pandas numpy matplotlib sentence-transformers
```

Run all cells in 01-kaggle-experiment.ipynb top to bottom. All outputs are saved to results/.

References

[1] Fraser-Taliente K., Kantamneni S., Ong E. et al. (2026). *Natural Language Autoencoders Produce Unsupervised Explanations of LLM Activations.* Transformer Circuits Thread. https://transformer-circuits.pub/2026/nla/index.html

[2] Allal L. et al. (2024). *SmolLM2 — with great data, comes great performance.* HuggingFace.

[3] Eldan R. & Li Y. (2023). *TinyStories: How Small Can Language Models Be and Still Speak Coherent English?*

[4] Chung H. W. et al. (2022). *Scaling Instruction-Finetuned Language Models* (Flan-T5).

[5] Reimers N. & Gurevych I. (2019). *Sentence-BERT: Sentence Embeddings using Siamese BERT-Networks.*